

Joseph Farkas

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science, Devices and Intelligence

Minors in Robotics and German. Concert Orchestra Member

Atlanta, GA

Expected May 2027

GPA: 4.00

- Relevant Coursework: Data Structures and Algorithms, Objects and Design, Robotics and Perception

PROFESSIONAL EXPERIENCE

Autonomous Systems Lead, Software Mentor

North Gwinnett Robotics

Oct 2021 – Present

Suwanee, GA

- Mentoring 15 high school students in sensor fusion and control using C++ for 115lb competition robot
- Integrating computer vision localization with on-the-fly A* pathfinding to power fully autonomous scoring
- World Comp. Autonomous Award & Division Champions. **Rated #1 globally for autonomous performance**

Research Assistant

Georgia Tech Engineering Space Policy Lab

Jan 2026 – Present

Atlanta, GA

- Modeling the downrange risk of space launches to inform policy decisions on launch safety and risk mitigation
- Scraping FAA NOTAMs to build a dataset of launch attempts; Plotting risk contours using QGIS

Software Intern

Viasat Inc.

Aug 2024 – May 2025

Duluth, GA

- Developed C/C++ firmware and Typescript frontend for prototype three-axis satellite antennas
- Built algorithms for analyzing signal quality and atmospheric modeling; integrated CI pipelines for validation

UX Assistant

Georgia Tech PACE High Performance Computing Center

Aug 2025 – Present

Atlanta, GA

- Supporting roughly 5,000 researchers across high-performance computing systems with incident triage
- Handling 30+ tickets weekly, resolving scheduler and accounting issues with a 90%+ resolution rate

ACTIVITIES

Guidance, Navigation and Control Team

Propulsive Landers @ Georgia Tech

Aug 2025 – Present

Atlanta, GA

- Applying PID+LQR, nonlinear MPC and lossless convexification to optimize rocket's fuel in powered descent
- Implementing Chebyshev-Gauss-Lobatto nodes for faster on-device trajectory replanning in Rust
- Prototyping with Monoprop UAV (drone); Aiming for 60m hop with 1000N rocket by EOY 2026

RoboCup: Autonomous Software Team

RoboJackets @ Georgia Tech

Aug 2025 – Present

Atlanta, GA

- Building multi-agent pathfinding, obstacle avoidance and task allocation in C++ for 6v6 robot soccer

PROJECTS, PRESENTATIONS & HACKATHONS

Evaluating Numerical Solving Methods for Optimal Powered Descent

AIAA Region II Student Conference

Mar 2026

Columbia, SC

- Compared ZOH discretization and pseudospectral parameterization to plan trajectories for a 3 DOF rocket lander
- Evaluated error from optimal solution at various solver resolutions for fast on-the-fly replanning

Holonomic Auto-Alignment using CV Localization | C++, Java, ArUco

Applied second-order kinematics and jerk limiting to replicate S-curve profile for discretized holonomic drive

Aug 2023 – May 2025

- Fused CV estimates from fiducials with filtered odometry using a state-space model for real-time robot location
- Allowed competition robot to align to any scoring position with < 1cm accuracy regardless of starting state

Foundations of Autonomous Programming | Georgia FIRST Robotics Symposium

Presented at statewide symposium on methods for teaching autonomous programming to high school students

Sept 2024

TECHNICAL SKILLS

Computer Languages: Python, Java, C, C++, TypeScript, Rust

Frameworks & Libraries: TensorFlow, YOLO, Jupyter, ROS, Matplotlib, OpenCV, ArUco, React